SentinelVision AI

Volume 09

Testing Documentation

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> **SentinelVision AI — Enterprise Documentation Suite**

> Status legend: Implemented | Planned | Future

1. Test Strategy

Follow the test pyramid: unit -> integration/API -> E2E -> performance/security/UAT.

2. Current Baseline

Suite	Count	Status
Backend API tests	5	Implemented
Edge tests	2	Implemented
Frontend tests	1	Implemented

Coverage target for core backend: >= 80% (Planned enforcement).

3. Test Environments

Env	Data	Usage
Local SQLite/Postgres	seed_dev_data	Developer
CI ephemeral	migrations + factories	PR validation
Staging	anonymized	Integration/E2E

4. Test Case Catalog

ID	Test	Status	Requirement Ref
TEST-API-001	Health endpoint returns healthy	Implemented	FR-DASH-01
TEST-API-002	JWT login returns tokens	Implemented	UC-AUTH-01
TEST-API-003	Tenant camera isolation	Implemented	BR-TEN-01
TEST-API-004	Event ingest creates event	Implemented	UC-EVT-01
TEST-API-005	Duplicate edge_event_id idempotent	Implemented	UC-EVT-01
TEST-API-006	Dashboard summary metrics	Implemented	FR-DASH-01
TEST-EDGE-001	Simulated detector structure	Implemented	AI-TRACK
TEST-EDGE-002	Event payload builder	Implemented	API-EVT-INGEST

ID	Test	Status	Requirement Ref
TEST-UI-001	API base defaults to /api/v1	Implemented	UI-DASH
TEST-API-007	RBAC denies viewer config write	Planned	FR-CAM-01
TEST-API-008	Cross-tenant ingest forbidden	Planned	BR-EDGE-01
TEST-E2E-001	Login to alert acknowledgement	Planned	UC-ALERT-01
TEST-AI-001	YOLO precision on validation set	Planned	AI-FACE
TEST-PERF-001	Ingest 100 events/sec soak	Planned	NFR-01
TEST-SEC-001	OWASP API top 10 scan	Planned	NFR-05

5. Unit Testing

- Django models, serializers, permissions, alert rule evaluation
- Edge inference helpers and payload builder

6. Integration/API Testing

- DRF APIClient with JWT and edge headers
- Tenant isolation across all list endpoints (Planned expansion)

7. E2E Testing (Planned)

Playwright flows: login, dashboard load, event review, alert acknowledgement.

8. AI Validation (Planned)

- Fixed benchmark clips per module
- Precision/recall/F1 reporting
- False positive rate monitoring in staging

9. Performance Testing (Planned)

Locust scenarios for event ingest and dashboard read load.

10. Security Testing (Planned)

- Bandit static analysis
- Auth bypass attempts
- Tenant isolation fuzz tests
- Pen test before production

11. UAT

Entry criteria: staging deploy green, P0 test cases pass, known defects triaged.

Exit criteria: stakeholder sign-off on intrusion/alert workflows.

12. Defect Severity

Level	Definition
S1	Production down or security breach
S2	Major feature broken, no workaround
S3	Minor feature issue with workaround
S4	Cosmetic/documentation